

Kinds in Haskell and Their Analogues in F# and JavaScript

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Abstract

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1 Kinds in Haskell

The functional programming language Haskell offers besides a classical typesystem a "typesystem of a typesystem", also called *kinds* (i.e. kind of a type or type of a type). In GHCi, the interactive haskell interpreter, we can use the ":t" command to check the type of a previously declared variable:

```
ghci> x = 42 :: Int
ghci> :t x
x :: Int
```

Now, the type "Int" has the following kind, which we can check using the ":k" command:

```
ghci> :k Int
Int :: *
```

In contrast, we can check the kind of a type constructor such as "Maybe":

```
ghci> :k Maybe
Maybe :: * -> *
```

2 Kinds and types in JavaScript

3 Kinds and types in F#

4 Conclusion

A Supplementary Material